

Title: Towards a neural exposome of India

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Background: India's age 60+ population is expected to double to 300 million people by 2050, forecasting a dramatic increase in dementia cases. Nearly half of these cases are expected to be driven by modifiable dementia risk factors (MDRFs). However, little is presently known about India's neural exposome, i.e. how the country's MDRF exposures relate to brain pathology outcomes. Here we make progress on revealing this neural exposome by associating a comprehensive set of MDRFs (including all 14 factors identified by the 2024 Lancet Commission report) with MRI-based neurodegeneration measures in older Indians who are either cognitively unimpaired (CU; CDR 0) or have questionable/mild cognitive impairment (CI; CDR 0.5/1).

Methods: From Wave 2 of the Longitudinal Aging Study in India-Diagnostic Assessment of Dementia (LASI-DAD), 602 participants were used. These participants had basic demographic information (age, sex), 16 MDRF measures (socioeconomic, health, environmental) and five Freesurfer-based measures derived from 3T T1-weighted MRI (Table 1). As MRI was collected at 11 sites, all MRI-based measures were harmonized using NeuroCombat, removing site variance while preserving variance attributable to age, sex, cognitive status (CU vs CI) and head circumference. For each pair of MDRFs and MRI-based measures, we built a Poisson regression with robust covariance to model the relative risk (RR) of excess atrophy in a given MRI-based measure [at least 1 SD below prediction from a normative model with age, sex, total intracranial volume (TIV) and time between non-MRI and MRI measurements (time gap)] as outcome and a given MDRF as exposure. All models controlled for age, sex, TIV and time gap (Figure 1).

Results: Obesity (BMI, waist circumference) decreases the RR of excess hippocampal atrophy while social isolation, PM2.5 air pollution and poverty increase the RR (Figure 2a). PM2.5 air pollution further increases the RR of both lower temporal lobe and lower whole cortical thickness (Figure 2b). No other associations between MDRFs and MRI-based measures (volume, thickness) survived FDR correction.

Conclusions: Obesity (in the unexpected direction), social isolation, PM2.5 air pollution and poverty may have early influences on neurodegeneration, especially hippocampal atrophy and decreased cortical thickness, among globally established MDRFs in India.

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Table 1: Demographic, modifiable risk factor and MRI measures from LASI-DAD Wave 2 neuroimaging cohort.

Demographics		Regional brain volumes	
n total	602	Hippocampal volume (mm³)	7362.4 (831.1)
Age, years	69.3 (6.5)	Abnormal: n (%)	108 (18%)
Sex, n female (%)	319 (53%)	Normal: n (%)	494 (82%)
Clinical Dementia Rating (CDR)		Temporal lobe volume (mm³)	78878.7 (8442.1)
0: no impairment: n (%)	355 (59%)	Abnormal: n (%)	122 (20%)
0.5: questionable impairment: n (%)	238 (40%)	Normal: n (%)	480 (80%)
1: mild impairment: n (%)	9 (1%)	Whole cortical volume (mm³)	377265.8 (36618.8)
Modifiable Risk Factors		Abnormal: n (%)	124 (21%)
Educational attainment		Normal: n (%)	478 (79%)
Never attended school: n (%)	273 (45%)	Total intracranial volume (mm³)	1318940.7 (158813.1)
Some schooling: n (%)	329 (55%)	Cortical thicknesses	
Urbanicity		Temporal lobe thickness (cm)	2.7 (0.1)
Urban-living: n (%)	347 (58%)	Abnormal: n (%)	110 (18%)
Rural-living: n (%)	255 (42%)	Normal: n (%)	492 (82%)
Hearing loss		Whole cortical thickness (cm)	2.4 (0.1)
Moderate/severe: n (%)	229 (43%)	Abnormal: n (%)	105 (17%)
None/some: n (%)	307 (57%)	Normal: n (%)	497 (83%)
LDL Cholesterol (mg/dL)	113.0 (37.6)		
≥ 160 mg/dL: n (%)	54 (9%)		
< 160 mg/dL: n (%)	548 (91%)		
Depression, CESD score (units)	9.4 (5.1)		
≥ 16 score: n (%)	70 (12%)		
< 16 score: n (%)	532 (88%)		
Head injury, ever had severe injury			
Yes: n (%)	80 (13%)		
No: n (%)	522 (87%)		
Physical inactivity*			
No moderate/vigorous exercise: n (%)	102 (26%)		
Some moderate/vigorous exercise: n (%)	297 (74%)		
Diabetes: HbA1c (%)	6.6 (1.6)		
≥ 6.5%: n (%)	165 (27%)		
< 6.5%: n (%)	437 (73%)		
Smoking*			
Currently smokes: n (%)	75 (19%)		
Does not smoke: n (%)	323 (81%)		
Hypertension, systolic BP (mmHg)	136.8 (23.3)		
≥ 140 mmHg: n (%)	239 (40%)		
< 140 mmHg: n (%)	363 (60%)		
Obesity: BMI (kg/m²)	24.5 (4.9)		
≥ 25 kg/m ² : n (%)	248 (41%)		
< 25 kg/m ² : n (%)	354 (59%)		
Obesity: waist circumference, female, male (cm)*	89.7 (12.5), 90.3 (13.9)		
≥ 88 cm female, ≥ 102 cm male: n (%)	57 (10%)		
< 88 cm female, < 102 cm male: n (%)	542 (90%)		
Excessive alcohol*			
Drinking 5+ days/week: n (%)	9 (2%)		
Drinking less frequently: n (%)	389 (98%)		
Social isolation, index (units)	6.6 (2.4)		
≥ 8 out of 12 units: n (%)	140 (35%)		
< 8 out of 12 units: n (%)	256 (65%)		
Air Pollution: PM2.5 (g/m³)	49.1 (22.4)		
≥ 40 g/m ³ : n (%)	329 (55%)		
< 40 g/m ³ : n (%)	273 (45%)		
Vision impairment*			
Moderate/severe: n (%)	88 (24%)		
Some/none: n (%)	279 (76%)		
Poverty, -log₁₀ CPERC (-log₁₀ rupees)*	-4.6 (0.2)		
At or below poverty line: n (%)	25 (6%)		
Above the poverty line: n (%)	376 (94%)		

Mean (SD) are provided for the following measures: age, LDL cholesterol, depression (CESD), diabetes (HbA1c), hypertension (systolic BP), obesity (BMI and waist circumference), social isolation (social isolation index), air pollution (PM2.5) and poverty [-log₁₀ consumption per capita (CPERC)]. All measures were acquired in LASI-DAD Wave 2 (2022-2024) unless marked by (*). Measures marked by (*) were acquired at LASI Wave 1 (2017-2019) and are available in a subset of participants: physical inactivity (n = 399), smoking (n = 398), excessive alcohol (n = 398), social isolation (n = 396), vision impairment (n = 367) and poverty (n = 401). Hearing loss was the only incomplete measure acquired in LASI-DAD Wave 2 (n = 536). Mean (SD) is also provided for brain volume and cortical thickness measures. For these 'normal' indicates that the residualized measure (residualized using a normative model built with CN participants) has not decreased by at least one SD (i.e. no excess atrophy relative to normal) and 'abnormal' indicates that the measure's decrease exceeds one SD (i.e. excess atrophy).

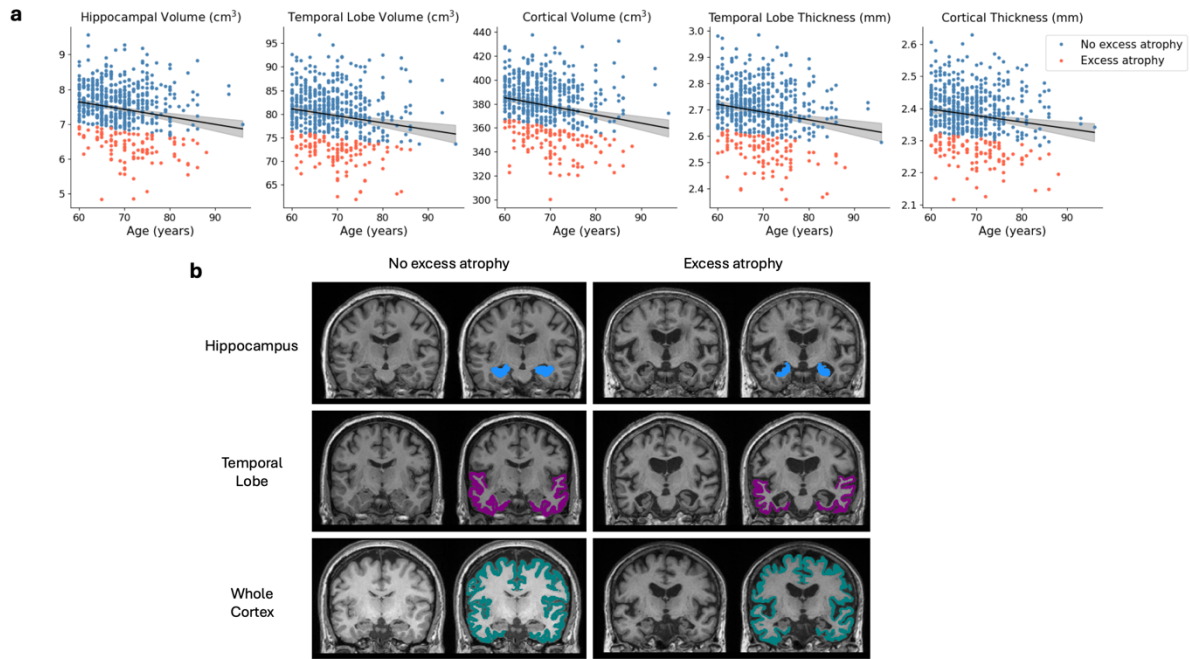


Figure 1 a. MRI-derived measures plotted against age. Red dots: those with residualized measures (residualized using a normative model built with cognitively unimpaired participants) at least one SD lower than expected (i.e. excess atrophy); blue dots: those with residualized measures not less than one SD (i.e. no excess atrophy). Trajectory of normal cognitive decline plotted as black line with gray CI. Binary excess atrophy status is the outcome in our models. **b.** MRI slices showing both the raw MRI slice and Freesurfer segmentations of hippocampus (volume), temporal lobe (volume and thickness) and whole cortex (volume and thickness). Each row compares one participant with no excess atrophy to one with excess atrophy. Participants are age and sex matched in each row.

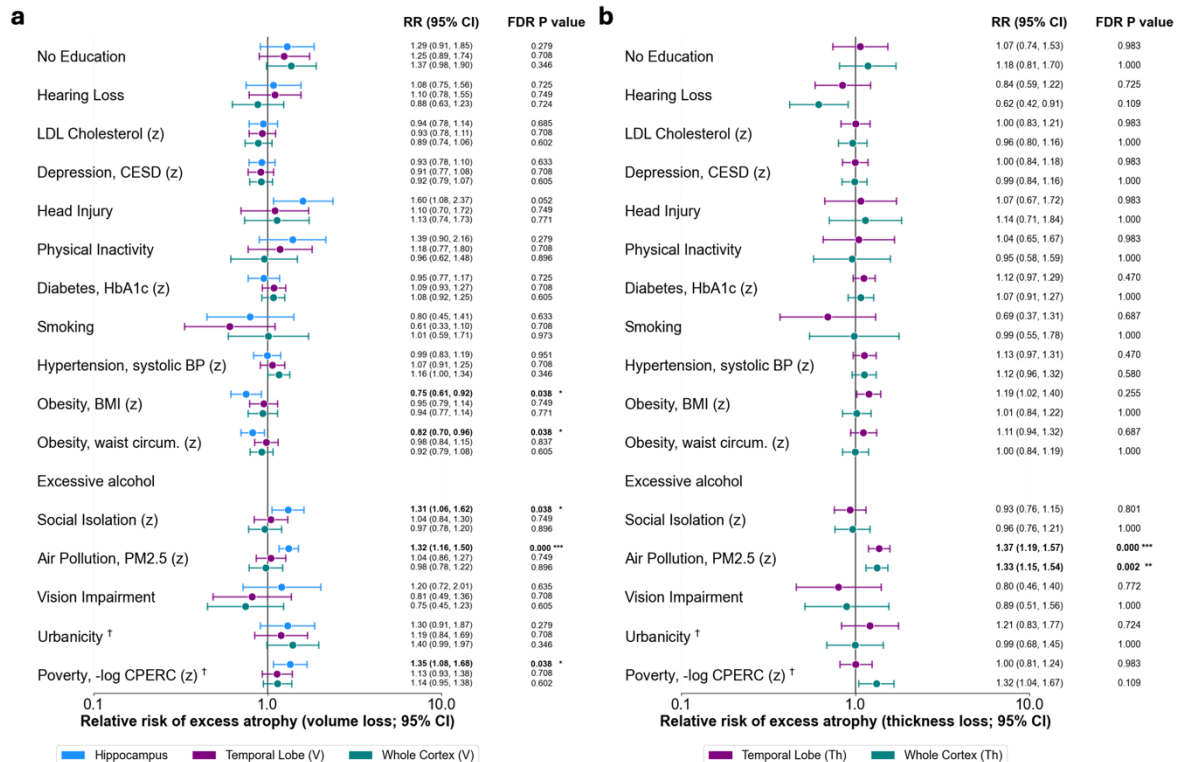


Figure 2 a. Forest plot of estimated relative risk (RR) of excess atrophy (decreased volume) in hippocampus, temporal lobe and whole cortex regions due to an exposure to each of the 14 MDRFs identified by the 2024 Lancet Commission, plus urbanicity and poverty. (z) indicates the MDRF is a continuous, z-scored exposure; all other MDRFs are binary exposures. Models were false discovery rate (FDR) corrected within each of the three volumetric measures separately. RR estimates surviving FDR correction are printed in bold. **b.** Forest plot of estimated RR of excess atrophy (decreased thickness) in temporal lobe and whole cortex regions, with separate FDR correction in each of the two regions. *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$.